

Project Demo Planning

OptiCrop: Smart Agricultural Production Optimization Engine

Date	28 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Project Demo Planning

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the OptiCrop project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S. No	Demo Section	Description	Duration (mins)	Responsible Member
1	Project Introduction & Problem Statement	Brief overview of the agricultural problem — inefficient crop selection — and how OptiCrop addresses it using ML.	3	Mondam Sandhya
2	Dataset & Preprocessing	Walkthrough of the crop dataset (N, P, K, temp, humidity, pH, rainfall), data cleaning steps, and feature engineering using Pandas and NumPy.	4	Bhavani Dunna
3	ML Model Training & Evaluation	Demonstration of model training (Random Forest, Decision Tree, SVM, Naive Bayes), accuracy comparison, and selection of the best model using Scikit-learn.	5	Mondam Spandhana
4	Live Crop Recommendation Demo	Live demonstration of the Flask web app — user enters soil and climate parameters, and the system recommends the optimal crop in real time.	5	Mondam Sandhya
5	Data Visualizations & Insights	Display of Matplotlib/Seaborn charts — heatmaps, distribution plots, and bar charts showing crop-environment relationships and model performance.	4	Mondam Spandhana
6	Conclusion & Future Scope	Summary of achievements, implementation rate, limitations, and future enhancements such as real-time IoT sensor integration and mobile app support.	3	Bhavani Dunna

Demo Flow Summary

Step	Activity	Notes
1	Introduction & Problem Statement	Nikhila introduces OptiCrop; highlights crop loss due to poor selection; states project objective — 3 mins
2	Solution Overview	Walk through system architecture: dataset → preprocessing → ML model → Flask UI; show tech stack — 4 mins
3	Live Feature Demonstration	Enter real N, P, K, pH, humidity, temp, rainfall values; show crop recommendation output + visualizations — 14 mins
4	Q&A Session	Address mentor/evaluator questions on model choice, accuracy, dataset, and scalability — 5 mins